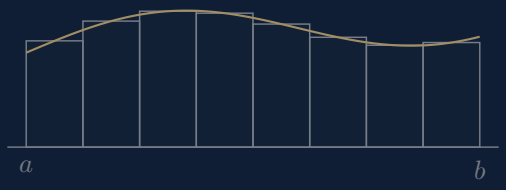
Mathematics Winter 2027

Real Analysis II

MATH 3162 · CRN 1964 · Undergraduate
MATH 4165 · CRN 1966 · Graduate

Analysis studies the continuum and functions through limits and approximation. It provides the foundations of calculus, with differentiation and integration among its central tools. This course develops integration, differentiation in several variables, and the convergence and approximation of functions. Discover how hidden functions emerge from calculus.



$$\sum_{j=1}^{n} f(\xi_j)\,\Delta x_j \longrightarrow \int_a^b f(x)\,dx \qquad (|\mathcal{P}| \to 0)$$

Topics

- Pointwise and uniform convergence of functions
- Regulated and Riemann integrals
- Fréchet differentiation
- Inverse and implicit function theorems
- Functional calculus on matrices

Instructor

Prof. Frédéric Latrémolière

Meetings

Tuesdays & Thursdays
12:00–1:50 p.m.
CMK 301

January 4–March 19, 2027

Prerequisite

Introduction to Real Analysis